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Requirements and challenges for implementing Design-based learning in kindergarten - - A case study in China

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Requirements and challenges for implementing Design-based learning in kindergarten—— A case study in China

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ABSTRACT

Design-based learning (DBL) is an inquiry-based learning method suitable to help children learn 21st century skills. It fits the way that children learn new things and skills. However, these novel teaching approaches can lead to challenges for teachers. So far, limited attention has been given to study teachers implementing DBL in kindergarten. To understand how tools can support teachers, it is important to understand what challenges they face. This paper presents an exploratory study involving two course designers collaborating with a formal-school teacher implementing DBL with 57 kindergarten students. From the study, we find that the challenges they perceived highly depend on their knowledge background. Thus, we introduce a framework called the DBL teacher's competency model and draw the connection between the knowledge required for teachers to implement DBL and the challenges teachers can experience in both course design and facilitating stages. This in turn provides support for designing educational tools or informing content for teachers' professional development.

CCS CONCEPTS

• **Applied computing** → *Collaborative learning*.

KEYWORDS

Design-based learning (DBL), kindergarten education, teacher competency, teacher's challenge

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1 INTRODUCTION

There is a growing interest in Design-based learning (DBL) in the field of education [29]. In design-based learning children learn by engaging in a project-based manner to address design challenges relevant to the learner. Similar to Cermak-Sassenrath and Møllenbach [7], we are interested in how design can be embedded in an educational approach. However, challenges exist in implementing DBL in practice. For example, teachers might have limited background knowledge about DBL and have difficulty transforming to new roles, such as being a facilitator, in DBL [18]. In previous work on developing tools to support teachers to implement DBL, we have examined teachers' challenges and opportunities in diverse contexts, like formal and informal learning, working with different target age groups, and within different educational systems. By looking at variations in contexts of use we hope to develop a better understanding of the requirements for providing good tool support for teachers. In a previous study, the challenges of teachers participating in an online training course about DBL were examined, based on observations and interviews with teacher trainers and student teachers [40]. A follow-up study regarding tool design was conducted, and we found that experienced teachers and inexperienced teachers can have different attitudes towards the same tool, and they might need a different kind of support. However, the manner in which the experience of the teachers differed was not explored in more detail.

Compared to the implementation of DBL in higher grades, implementing DBL in kindergarten has special requirements and corresponding challenges like students' limited relevant background and skills [3]. Tools might support teachers in overcoming some of the challenges of implementing DBL. However, little is known about the challenges and requirements of teaching DBL in kindergarten. It is important to develop a better understanding of educational practice, where the roles of teachers and students have changed from a more traditional practice, to know how tools can be designed to support the pedagogy of DBL [6].

In this paper, we describe a study that explores the challenges faced by teachers, and the requirements for teachers to implement DBL targeting a slightly younger age group in a formal context. Here we followed an exploratory participant approach to the case study, to develop a more in-depth understanding of the challenges of teachers in implementing (designing and facilitating) DBL activities.

The study was implemented in a public and in a private kindergarten in China.

We introduce a DBL teacher's competence model to describe and summarize our findings. Furthermore, we present requirements for tools that can support teachers in designing and facilitating DBL related to the competence model and discuss the possible improvement and extensions for it in the discussion and conclusion section.

2 BACKGROUND

2.1 Kindergarten education

Kindergarten education, as well as pre-school, is an important period for children to develop multiple skills, such as social skills [27]. It can be seen as part of early childhood education (ECEC), which includes the age range from zero to six years old [26].

The kindergarten education systems and the required competence for teachers differ for different countries and regions. In China, some children would enroll between the ages of four and six or even older in remote areas instead of 3-5 as suggested [27]. Kindergarten teachers need to learn and teach integrated fields including "Health, language, society, science, art", which is different from primary education in which the teachers are more focused on the specific disciplines they are teaching. In recent years, the kindergarten textbook changed to theme-based units with all subjects instead of discipline-based with different topics. Thus, to help children learn knowledge in a more integrated way.

In Europe, such as in Finland, kindergarten teachers tend to place a strong emphasis on child-centered activities, and play has an important role in this age [24, 38]. Public kindergartens are free of charge in Nordic countries such as Finland and Denmark to give every child the opportunity to attend school. The teaching profession is highly valued in Finland, and there are common guidelines for educational activities in classrooms of all levels, these ensure the schools in Finland have fewer differences and keep high quality [24]. In the Netherlands, various kinds of preschool organizations exist. The daycare center and preschool are targeting 2–4-year-old children. Both in preschool and primary school, teachers are responsible for all subjects [26].

Particularly, national curricula do not yet exist in the Netherlands at any level of education for this age group, so the school and the teacher can have the autonomy to choose what is needed for children and what to teach [31]. In 2000, they started a program called "Early Childhood Education" (in Dutch: VVE-beleid) for children with language disadvantages from age 2.5-6.

In general, kindergarten teachers and students have diverse knowledge backgrounds in different countries and school systems, which might influence their experience and challenges when implementing DBL in their context.

2.2 Competence model for teachers

A teacher trying to implement DBL in an educational setting might run into several challenges that are contingent upon and can be influenced by their experience and knowledge. A general model to describe a teacher's knowledge is the pedagogical content knowledge model (PCK) by Shulman which includes three elements [25].

Subject matter content knowledge depends on which subject-related area it is, and the way of discussing the content structure knowledge differs. Content knowledge refers to the body of knowledge and information that teachers teach, and the students expect to learn about the subject.

Pedagogical content knowledge includes examples, explanations, and other ways of representing the subject to make it more comprehensible.

Curricular knowledge is like the root or body structure of a discipline or field, once we have the structure, then we can attach content like leaves grow on a tree.

Different models for describing a teacher's knowledge exist including those developed based on Shulman's model, such as the Technological Pedagogical Content Knowledge (TPACK) model which embedded a technology factor in the previous framework [15].

2.3 Design-based learning and DBL-alike approach for early-childhood

In many countries, education is changing to meet future challenges. So-called 21st century skills, such as critical thinking, problem-solving skill, and creativity, are identified as critically important [29]. A Design-Based Learning approach can support in developing of such 21st century skills. Design-based learning (DBL) is also called learning by design [17]. DBL is used to teach integrated STEM and 21st century skills through various embedded activities during the course [39]. DBL shares the core of an inquiry-based process with other methods like Problem-based learning (PBL) and Design Thinking [2]. Compared to other inquiry-based pedagogies, DBL focuses more on design methods and emphasizes hands-on and mind-on activities [40].

DBL can be used from kindergarten to higher education to teach the fields. The abilities of kindergarten teachers and children vary much through countries and regions. They might lack the skills to implement DBL or lack interdisciplinary context knowledge they want to teach, such as integrated STEM [11].

In some European countries such as Germany and the Netherlands, the federal level of the curriculum is not implemented in kindergarten [37]. Some kindergartens interested in a project and design–based approach choose to cooperate with the organizations like "Little Scientists' House" (in German: Haus der Kleinen Forscher) which provides a teaching package for early childhood education working towards real-life problems [14]. In the US, Resnick argues the so-called "kindergarten approach to learning" which is characterized by a spiraling cycle of Imagine, Create, Play, Share, Reflect, and back to Imagine. This model shares a similar process with DBL [22]. There are also informal organizations that provide resources like STEAM experiment packages for preschool teachers and children to work towards problem-solving [21]. In China, several informal organizations provide DBL courses or games for kindergarten-age students and cooperated with some formal schools in both cities and remote areas [30].

2.4 Challenges in teaching DBL and tools supporting teachers

In previous studies, challenges teachers face include sustaining learners' motivation, accessibility of investigation techniques, background knowledge of students, management of extended activities,

and the practical constraints of the learning context [10, 23]. A more recent study conducted during a teacher education course including a facilitating session found that teachers experienced much more challenges during the facilitating phase than receiving the teacher education course. The challenges teachers perceived include time management, managing children's teamwork, asking and answering open questions, and keeping the children focused [40]. In DBL, the role of the teacher shifts more towards being a facilitator. A teacher, therefore, needs to know how to manage the timing, introduce knowledge, and help students reflect appropriately [16]. Understanding the challenges teachers meet can be the basis for designing tools for teachers.

As for tools designed to support teachers implementing DBL, they are mainly used in two phases, one is for designing the course, and another is for facilitating the course. The challenges teachers meet, and the tools needed differ between the two phases. From traditional kinds of teaching to DBL, the courses change from teacher-centered to student-centered, and from knowledge-based to inquiry-based. Furthermore, in DBL, teachers play the role as coach and facilitator, giving scaffolding when students need support exploring the open-ended questions [35, 41]. Some teachers implement DBL based on well-designed courses and materials provided by various educational organizations (such as the d.school and the Designathon approach [4, 9]) so that they might only need support when they run and facilitate the course. Others are designing the course by themselves, so they might need tools and support for both the designing and facilitating phases of the course. There are some existing tools to support teachers in implementing Design-based learning, such as the commercially available GROOW toolkit and the Teach21 Online Authoring Tool which both support curriculum design [28, 29]. In a study with primary school teachers during the course designing phase, there are some insights about the tool or material design. For example, the course designing materials need to be transferable because DBL courses are mostly designed by a group of teachers rather than an individual teacher. Though those teachers have some experience with DBL and ME (Maker Education), they still think further explanation about DBL activities is helpful because they only used several activities before, and with support, they can get to know new activities in a short time and implement [36]. Thus, instead of defining the participant as experienced or inexperienced, there should be different levels of experience for participants, and this influences their needs for tools. Since the openness and diversity of a DBL course and process, and although various tools and materials exist, teachers still have difficulties implementing DBL.

3 CONTEXT OF THE STUDY AND RESEARCH METHODS

In a previous study, we gathered information about teacher experiences with preparing and facilitating DBL through interviews with multiple teachers. In the previous study the teaching material was provided by an organisation called Designathon works, so teachers did not need to design the course by themselves. The challenges teachers felt were mainly during facilitating [40]. The present study builds on this work by gathering rich and in-depth information through a participant observation case study where the researchers co-designed the DBL materials with teacher. The study focused on the following research questions:

1) What are the requirements needed to design educational DBL materials and facilitate DBL in kindergarten?

2) What is challenging about designing and facilitating DBL in kindergarten?

A qualitative research approach was used in this study to explore the possible factors and develop a deep understanding of these processes [1, 8].

3.1 Background of participants and kindergarten context

As shown in Table 1, participants in this study include a formal kindergarten teacher and two course designers. The teacher has 28 years of kindergarten-teaching experience but has neither experience with DBL-alike approaches nor knowledge about Design Thinking. She is willing to implement the new pedagogy to teach and shows enthusiasm. The course designers have a design background and are knowledgeable about the Design Thinking method, which shares a similar process with Design-based learning pedagogy [12]. But the course designers have neither experience teaching DBL-alike methods nor teaching in a kindergarten. The first author of this paper participated in the study as one of the course designers.

The present study that explores the role of the participants' background might seem like an extreme context as problem-based learning approaches are widely adopted in some countries such as Germany. But teachers in Chinese public schools normally have much less experience with DBL-related approaches because of different educational system.

Besides, two classes of children aged 5-6 from different kindergartens participated in the study, both kindergartens were in a medium-sized city in Sichuan province, China. The number of students from private and public kindergarten is 18 and 39, respectively, with 2-3 teachers in each class. Thus, the amount of students per teachers is more than twice as much in public than in private kindergartens.

As for the topic, we chose the communication tools, the two kindergarten students just learnt this topic based on the curriculum at school and the course content is related to multiple disciplines including Art, Science, and Society. Students have some knowledge about the chosen topic (design challenge), but not about the DBL approach (see Table 2). In recent years, the kindergarten does not formally teach Chinese spelling in public kindergarten, to lighten the children's burden. Thus, verbal instruction from teachers becomes more essential during the course setting.

3.2 Process to design and facilitate the course

Because of the societal situation (the Corona pandemic), the course designers and the teacher are doing the whole course preparation process online. Only the course facilitation is offline with two classes of children in separate kindergartens.

The overall process of course design and facilitating DBL in two classes included (see Table 3):

1. Course design (preparation)

Table 1: TEACHER AND COURSE DESIGNERS' BACKGROUND

Role	Background/Experience		
	Teaching experience	DBL/ Design Thinking	Topic-related knowledge
Teacher	Yes	No	Yes
Course designer A	No	Yes	Yes
Course designer B	No	Yes	Yes

Table 2: STUDENTS' BACKGROUND

Student	Background/Experience	
	Design-based learning	Topic-related knowledge
Class A	No	Yes
Class B	No	Yes



Figure 1: Two classes of kindergarten children during the "Ideate" phase (Left: private kindergarten; Right: public kindergarten)

- Discussion about the topic and general time setting of the DBL course.
- Design the course structure and slides by the course designers and send them to teachers for feedback.
- Design the template of the children's worksheets.
- Course designers write the course description and guidance including the learning goal and possible difficulties, description for every slide, and usage of the template

2. Facilitate and use DBL materials in class (see Table 4).

3. Reflect on the quality of the course materials by course designers and teacher.

4. Improve the design of the course, and

5. Facilitate in another class.

Apart from the basic DBL process, designers also used the curricular spiderweb to decide on educational design decisions. It includes the basic elements such as learning objective, content, activities, teacher role, etc., which are summarized from the general teaching context which teachers need to consider when implementing a course [20].

The topic and design challenge chosen is designing for better communication between the children themselves and the people they miss most but cannot meet during staying at home. On the one hand, it is based on the scenario related to the children's experience, because the kindergartens had been suspended for one semester in 2020 before the class. On the other hand, the two classes had learned the topic of communication tools, which is one of the themes in the upper-grade kindergarten curriculum (totally 3yrs).

After that, we designed the course material, including slides teachers use in class and lesson plans including instructions for every slide. Then, we designed the templates of the worksheets for children to use in the Ideate and Show phase. The course designers created the main materials, and the teacher provided feedback on the materials.

3.3 Process of data gathering during the study

Based on the research questions, of **what are the requirements related to teaching DBL** and **what is challenging in designing and facilitating DBL** in kindergarten, data was collected in the study, as shown below:

1. Course design diary filled in by course designer A includes documentation of the online iterative process and discussions between teacher and course designers.

2. Videos during facilitating the course.

3. Filled templates and drawings made by the children gathered during the course.

4. Semi-structured interviews with the teacher and course designers after both courses.

3.4 Data analysis of the study

During the study, open coding, axial coding, and selective coding are used [8]. Open coding is normally the first step. It is an interpretive process by breaking down data analytically to form categories and subcategories. Secondly, the axial coding includes connecting categories to subcategories. The last step is selective coding. All

Table 3: Course implementation

Class	Stage	Place	Main role	Minor role
Class A	Course design (Preparation)	Online	CD	T
	Facilitating	Classroom	T, S	/
	Reflect on course	Online	CD, T	/
Class B	Improve the course (Preparation)	Online	CD	T
	Facilitating	Classroom	T	/

CD=Course designers, T=Teacher, S=Students[1]

Table 4: Phases of the facilitating stage

Course	Phases during Facilitating	Explanation	Materials used
Day1	Empathy	Tell the story based on the topic	Slides
	Define	Define the problem	Slides
	Individual Ideate	Think about the problem based on the given scenario meet by themselves	Template for individual ideation
	Show	By recording, telling the story about the individual ideate	Phone, template from the previous phase
Day2	Group Ideate	Think about ideas as a group and choose one to work on, Imaging the details of the idea chosen from the group ideate	Template for group ideation and sketch
	Reflect	Formative-reflect during the course	Prepared questions

the categories are unified around "core" categories which can build on a grounded theoretical framework [5, 8].

In this study, the data are analyzed using the following processes:

1. Transforming the audio and video data to text with pictures.
2. Based on the research question, coding interesting features of the data regarding the entire data set and collating data relevant to each code.
3. Sorting out the codes and finding potential connections, gathering the data relevant to the potential sub-categories.
4. Checking the sub-categories and combining them to form a thematic "map" based on the generated theoretical framework.
5. Defining and rechecking the names of the categories, generating clear definitions for each category.
6. Finding proper examples and argumentation from both the original data set and other related research, making the thematic "map" clear to target readers.

4 RESULTS

Based on the study we present three types of results: a proposal for a framework of teacher's competencies in a DBL context, requirements for teaching DBL, and challenges of teaching DBL in Kindergarten. The framework itself bases on both coding process and previous model by Shulman called teacher's general content knowledge.

4.1 Proposed framework for teacher's competency in the DBL context

From the study, we find that the challenges participants faced in the course design and facilitating stages highly depend on their background knowledge. The coding showed that the types of challenges experienced by teachers with a lot of teaching experience, but little DBL experience are very different than the challenges experienced by the participants that have less general teaching experience but more experience with DBL. Thus, we propose an adjusted version of Shulman's generic PCK model, that is specific to a DBL context (see Figure 2). The adaptation is based on linking the thematic map from this study and the general teacher's knowledge model mentioned in 2.2 [25].

Through iterative coding, we find that the challenges perceived are related to what kind of knowledge or experience course designers and teachers have. Meanwhile, we find that regarding Shulman's PCK model, DBL has some special focus including learning through design instead of content knowledge-based, the knowledge needed for teachers should also be different. Thus, we proposed a framework called the DBL teacher's competence model which is transformed from the teacher's general content model by Shulman to fit the DBL context as shown in Figure 2.

DBL-related knowledge

DBL-related knowledge relates to both **pedagogical content knowledge** and **curricular knowledge**; it indicates teachers' knowledge of the DBL processes and practices [16]. DBL always

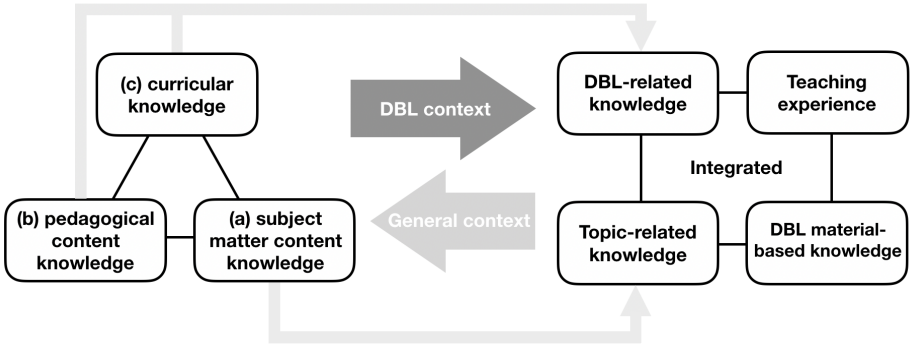


Figure 2: From the teacher's general content knowledge [18] model (on the left) to the DBL teacher's competence model (on the right)

includes the phases: introduce a design challenge, explore the possibilities and solution, cooperate to draw, make, reflect, and show. It may have different names from other studies and projects, but the core is to work towards problem-solving and explore by design instead of remembering fixed concepts.

Teaching experience
Teaching experience does not directly belong to the three elements, but Shulman mentioned a related concept called **general pedagogical knowledge**. On the one hand, it includes the knowledge of target learners and their background, principles of school organization, finance and management, and other socio-cultural aspects [25, 33]. On the other hand, it also includes methods of teaching.

Topic-related knowledge
Topic-related knowledge is related to **subject matter content knowledge**, it is teachers' knowledge about the subject to be learned or taught [15, 25]. But for DBL, there is no boundary about the subject, but the approach is problem-driven and always interdisciplinary.

DBL material-based knowledge
DBL material-based knowledge has no direct relationship with the general content knowledge model. It indicates the specific materials used during Make or any other phase in which the children do hands and minds-on activities. Those materials are not necessarily with technology and electricity but can be also paper-based or waste materials-based depending on the course setting and resources teachers have.

4.2 Requirements related to implementing DBL

Through thematic coding, we find that the requirements can be described using 4 main themes because the challenges perceived by participants are exactly related to the background they have. DBL-related knowledge includes knowledge about the DBL process and some features and key elements regarding DBL.

4.2.1 Design-based learning-related requirements.
The basic course structure in this study is shown in Table 4, but the process can be adapted based on the teaching context and teacher's preferences. As for the learning goals, they can be set based on curriculum, 21st century skills-based, or other specific goals added by the teacher. In this study, we set the two main types of learning goals. The first type is related to 21st century skills including cultivating creativity, cooperation, and communication, and the second type is related to the topic in the curriculum, to help the children understand more about communication from different angles. During the course design step, the course designer needs to get familiar with the process and add the following content to the framework based on the learning goal.

To ensure a common understanding of the course design between the course designers and teacher, teaching guidance needs to be made as the teacher used in her daily practice, it includes explanations about the learning goals, materials, and possible difficulties, details of the activities, and explanation of the slides. The teaching guidance addresses typical course design elements and can be used before the class to get familiar with the whole course and during the facilitating process as a reminder for the teacher [20, 32].

Openness is a core DBL feature: there is no fixed answer when working towards the target problem [13]. Different answers are accepted in DBL. When interacting with students, the teacher needs to ask open questions: e.g., "What do you feel...?", "How do you...?", and "Why?". Simple closed questions that the students only need to be answered with "Yes" or "No" are not suggested but can be combined with open questions to let the students think and reflect more on their decision. Both the course designer and teacher need to be aware of the openness feature to make the open-ended problem-solving process smoother.

During teamwork, worksheet templates are a vital part. It is often used when children take the lead and make their designs. Depending on the activities embedded in the course, the template for DBL might break a task into small steps and include instructions that help the children to think. Templates are always used in the "Draw" and "Make" phases to get more ideas about the solution and to further develop it. Those parts of questions are always included: "Design for whom? What kind of problems do they meet? Can you imagine what it looks like?". When designing the template, children's development level also needs to be taken into consideration,

Table 5: DBL-related requirements

Keywords	Examples
DBL process	1. Basic structure needed
	2. Learning goal-based
	3. The design process is iterative
Openness	1. Different answers accepted
	2. Ask open questions instead of closed questions
Teamwork	1. Worksheet template design: Elements for the template used in the ideate phase, consider background level
	2. Dialogue and interaction: teacher &student, peer &peer
Time management	1. The time setting is open, and highly based on the teacher's experience
	2. The working time is different for different children

for kindergarten children who are not able to read well, the template needs to be as simple as possible, with fewer words, and short explanations.

Apart from the previous issues, dialogue and interaction are essential in DBL, e.g., those happening between teacher and student, and peer to peer. In this study, we focus more on the teacher's dialogue and interaction with the student. In different phases, teachers have different dialogue types and strategies toward children. When children take the lead and design, the teacher mainly communicates with the small groups and individuals to deal with specific questions. When the teacher speaks to the whole class, the dialogue is more about general instruction. During the facilitating or even preparation phase, understandable scaffolding and prompts might be needed for the teacher.

Compared with a traditional way of teaching, time management is more flexible in the DBL teaching process. Firstly, the time setting highly depends on the teachers, they can choose to add more activities or extra sessions, and they can also decide the level of difficulties. Secondly, during the activities where the children take the lead, the time needed might be more variable during the course, different children might need a different length of time to finish. Furthermore, the notion of "finishing" is also quite open, depending on the learning goal of the course setting.

4.2.2 *Teaching experience-related requirements.*

Teaching experience indicates general teaching experience towards target-age children, it includes understanding children's background level, fun factors, and classroom strategy (See Table 6). These can help course designers and teachers better understand the target children.

Firstly, children's background and abilities need to be taken into consideration. What they have learned, influences the time needed and difficulty level such as the depth and broadness of the concepts.

Teaching experience is also related to knowing more about what is fun or not fun for students, especially for kindergarten children who are easily distracted if one thing is not interesting or makes no sense to them. Through the study, we found that the children are more focused on visual and tangible stuff such as videos and pictures rather than audio-only information including storytelling without visual information. They prefer topics and examples relevant to their life instead of more abstract topics especially those involving memorizing tasks.

In the classroom strategy, the teacher used some rules with gamification to get the children focused. Course designers need to think about children's possible feedback towards the elements when designing the course such as time setting, length of the course, learning goal, scaffolding questions, and difficulty level. The course setting needs to be adjustable based on the teaching experience and teacher's preference. Related to teaching experience, teachers can also gain more diverse facilitating strategies like asking and answering open questions.

4.2.3 *Topic-related knowledge requirements.*

Topic-related knowledge is the knowledge about the topic chosen for the course, it includes authenticity, understanding of the topic, and external support (See Table 7).

The topic selected is usually different between different courses, especially for DBL which the course is problem-solving-oriented and might be interdisciplinary. The topic can be chosen based on the curriculum, or from SDG2030 (Sustainable development goals) [19]. The first thing that needs to be taken into consideration is authenticity. Authenticity can be linked to two components. On the one hand, the topic and example need to be connected to the real world, on the other hand, they need to relate to the children's background and interests. Thus, the content of the course can be better understood by the children and be meaningful for them. In this study, the topic we chose is related to communication tool design during the lockdown.

During facilitating, the teacher needs to know how to ask proper open questions and how to guide children to ideate and design. Using authentic examples that make sense for children, the teacher with teaching experience at a certain age knows more about what they have already learnt, what might be difficult for them, and what might be interesting for them. This helps to decide the proper difficulty level regarding the topic.

External support is often required, including expert support and school support. The teacher might need help to tell a good story in a certain field and it will be helpful if there are experts to help with explaining difficult concepts regarding the topic. If a school or its headmaster agrees on the learning methods of DBL, they can provide the materials and other resources the teacher needs.

Table 6: Teaching experience-related requirements

Keywords	Examples
Children's background level	1. Take children's background into account 2. Create a suitable difficulty level for the target age 3. Ability needed for children
Fun factors	1. Use scenario-based, authentic examples 2. Use visual and tangible things
Classroom strategy	1. Make course setting adjustable 2. Use clear rules and small games to get children focused 3. Use a DBL-specific facilitating strategy

Table 7: Topic-related knowledge requirements

Keywords	Examples
Authentic	1. Use authentic topic setting 2. Use scenario-based, understandable examples for children
Understand the topic	1. Teachers need to understand the topic first 2. Decide on the topic with a proper difficult level
External support	1. Expert support might be needed 2. Provide school support 3. Consider parents' involvement

Besides, linking DBL to parents' involvement can also contribute to broadening children's perspectives using methods such as reading storybooks with children, visiting museums, asking more questions, interacting more with them, etc.

4.2.4 Material-based knowledge requirements. Material-based knowledge includes material choosing and its usage (See Table 8). Materials are related to the materials or technology selected for making. In a previous study with teachers [40], we found the need for those materials is mainly towards the "Make" phases.

Different materials have different kinds and difficulty levels, with or without technology. More complex and less complex technologies can be used. The making materials without technology include paper-based toolkits and Lego bricks. The making materials with technology include Arduino, micro-bits, and online tools like Scratch.

4.3 Challenges during designing and facilitating the DBL course

In this study, the teacher and the course designers took the two roles of designing and facilitating separately and cooperated online. Through thematic coding, we find that the challenges highly depend on the participants' backgrounds. Besides, the specific (cultural) context also matters, such as the high student-to-faculty ratio which means one teacher does not have much time to spend with each student [5].

We coded the challenges for course design and facilitating separately based on the DBL teacher's competency model. The cells marked in gray indicate the challenges related to the categories where the participants lacked competency most (See Table 9).

4.3.1 Challenges during course design.
Although the course designers have **DBL-related knowledge**, teaching experience was lacking. During course designing, course designer B was not sure whether the time setting is suitable for children or not, course designer A suggested having a rough timetable so that the teacher can adjust it during the classroom.

Because the cooperation was online, course designer B thinks the communication is not efficient, sometimes she was guessing because there might not have instant feedback. Course designer A thinks it might help if there is a tool for online DBL co-designing and everyone can work on it whenever have time. The two course designers think it will help if the teacher can participate more in the course designing process. The teacher told the course designers her daily teaching materials including slides are based on textbooks, all the slides are sharable between different teachers, and they can adjust the slides. She thinks clarity and shareability are important so that different teachers can understand the course content easily, and that course design from scratch might be an extra burden to some kindergarten teachers.

For the course designers, because of lacking **teaching experience**, the knowledge about target-age children was limited. Learning goal setting and measuring is hard, especially in the case that the course design and facilitating's role are separated, the agreements and deeper understanding of the learning goals are especially important because the whole course should be developed based on that. But the course designers find global learning goals might be vague sometimes and they can be broken down into sub-learning goals to be embedded into the course design process.

The activities and the examples need to be authentic to the children and scaffolding questions are needed. Those are hard for course designers without teaching experience, they don't know children's background level, or what is fun or attractive for kindergarten children. For example, the topic is related to communication tools in this study and the children already learnt this topic at school, but

Table 8: Making material/(technology)-based knowledge requirements

Phase	Examples/Explanation
Material choosing	Choosing the making material based on difficulty, time, and cost
Material usage	Get familiar with how it works and teach students, especially during the 'Make' phase

Table 9: Challenges experienced by course designers and the teacher

Competence category	During course designing	During facilitating
DBL-related	1. Proper time setting for the course 2. Challenges regarding online cooperation during course design	1. Open dialogue with children and scaffolding 2. Children lack cooperation skills 3. Scaffold the ideate phase 4. Facilitating strategy
Teaching experience	1. Learning goal setting and assessment 2. Proper activity setting 3. Choosing authentic/meaningful examples 4. Set scaffolding questions	1. Classroom management 2. Managing too many children
Topic-related	1. Topic choosing 2. Problem-based storytelling of the course	1. Teachers need to familiarize themselves with the topic and instructions 2. Lack of expertise related to the topic
Making material-based	1. Lack of 'making' experience 2. Hard to select materials 3. Limited funding	1. Lack of making experience 2. Lack of external support 3. Failed to have agreed design to work on during 'Making' from the ideation phase

it is still not clear what they've learnt about this topic, and what cases are included in their textbook.

The course design process took 9 weeks including iterations with the teacher. The example chosen at first was expected to be too abstract and difficult for the children according to the teacher. In hindsight, both course designers think it would have been more effective if the teacher would participate in the course design in the early stage or that they experience several courses in kindergarten by themselves.

Regarding the **topic-related challenges**, topic-choosing, driving question setting, and problem-based storytelling are challenging: some topics and the way we address the logic and story were too abstract or made no sense for the target children. In this study, the course started with the authentic scenario that during lockdown that the children cannot meet the people they miss and do outdoor activities with friends. While the final driving question is to design communication tools for lockdown situations, this might be too abstract for the target children. One surprising thing was that some children didn't have negative feelings about staying at home as some adults felt: this arose in both classes. So, an alternative design challenge like marine protection might be understandable for children and less dependent on personal feelings. Furthermore, the same topic also can be set with different levels, for example, for younger children, we can choose easier and fewer examples, and a shorter course setting. Target children's feelings need to be considered regarding the course setting to make it more authentic.

For the **material-based challenges** part, the two course designers and teachers lack experience in certain making materials and are unsure about which might be suitable for kindergarten. Furthermore, funding for the materials is limited. The course designers searched for several options and discussed these with the teachers, but the teachers thought it would be hard for both teachers to teach and students to make. Also, the kindergarten cannot provide funding for trial lessons like in this study.

Overall, all four competencies are essential for course designers, and each will bring corresponding challenges if the course designer is not experienced in the competency.

4.3.2 Challenges during facilitating.

Regarding **DBL-related challenges**, for the teacher who has 18 years of teaching experience without trying DBL-related methods before, she still finds it hard to support open dialogue and scaffolding. Challenges for children and teachers are related to each other. The children lack DBL experience as well and have had limited opportunities to practice 21st century skills.

If there are some unexpected questions or comments from the children, the teacher might not know how to give further explanation. For example, one child asked what if he doesn't know how to draw, the teacher didn't give extra guidance but only ask him to try. For those children who haven't been taught reading at their kindergarten level, oral scaffolding is especially important. We find that the children might have a problem understanding long explanations or instructions: the teacher needs to make the explanation as short and clear as possible. In this study, as the teacher said, the children often asked what to do for the next step and had problems finishing the design part. From the video, we find that the children use random ways like "Rock-paper-scissors" to decide to work on



Figure 3: Individual and group ideation with the sketch (Left: drawing by the individual; Right: drawing by group)

which idea instead of discussing it. It might be difficult for children to take the lead and do design without practice. The children had limited experience with group work before, but she also mentioned that before the Ideate phase, the children already participated in the course for around 20 minutes, so they might be too tired to deal with challenging problems and make decisions.

As for the template used for generating ideas, the guide of a template set as reminders for children can be a limitation in some cases. For example, we drew a person's outline in the course about "Myself" but less than 3 out of 18 drew on the outline and others left it blank or drew a separate person after the teacher gave the explicit instruction. The course designers think it might be because the children think the outline does not look like themselves. This also reminded us that the examples and the template design should match the children's way of thinking, otherwise it will not work for children. In the future, course designers need to try out multiple ways to simplify the templates and see how children interact with them to make further improvements. Facilitating strategies for DBL also need to be developed that include strategies that cultivate cooperation and allocate individual roles for children. As you can see from Figure 4, children did well in the individual ideation, but they do not know how to work on the design together, all the children drew whatever they wanted instead of discussing and improving the common group design. Furthermore, the teacher was not sure how to guide the children in this process.

Even though the teacher has **teaching experience**, she still ran into challenges because of the differences between DBL and a traditional classroom. Firstly, unlike the traditional class in which the teacher takes a more central leading role, a large number of children in a class can bring difficulties implementing DBL. Ideally, the teacher needs to spend enough time with every group to know more about their ideas and encourage them, instead of having a fixed answer, as there is no wrong or right answer in DBL.

Secondly, DBL classroom management is also different compared to the traditional class. Take time management as an example: the time for children to brainstorm and make is uncertain and decided by the teacher. The teacher needs to use some strategies to make sure the time management is under control, if some children spend too much time drawing, then the teacher might tell the children that finishing is more important than being perfect.

For the **topic-related** part, familiarization with the topic and instructions was hard for the teacher, because of distant co-creation and less participation in the course designing phase. Although the teacher taught the topic regarding "communication tools" before, the logic and content of the "curriculum-based" course and DBL are quite different, so these are still new courses for the teacher to prepare. Besides, expert support and other external support might also be needed: for example, the teachers do not know how to explain examples like VR (Virtual reality) and other high-tech communication tools. The teacher thinks it can be very helpful if she can have the VR product for children to try, as tangible examples are always good for young children to learn new concepts.

The **material-based experience** is mainly related to the Make phase. Several digital materials options were discussed, but none of the participants had experience in using them. Both teacher and the course designers thought the Make phase would be challenging because of materials-choosing for kindergarten children and the funds needed. Teachers need a long time to get the budget to buy making materials, which the kindergarten might not support. The teacher has almost no experience teaching kindergarten children about making design models even though she has 18 years of teaching experience in both public and private schools in China.

In our previous study targeting 8-12 years old children in the Netherlands, course designers find that the Make phase has similar difficulties to the Ideate phase because children need to take control and make their own decision in the two phases [40].

Thus, course designers made two flexible options that the teachers can try during the Make phase: using plasticine or paper-based materials which they have in kindergarten, if the previous part especially the 'Ideate phase' went well, or if not, she can skip this part. Finally, the 'Make phase' was not implemented in this exploratory study. In this study, teacher training sessions were not provided. Participants in this study had diverse backgrounds and met different difficulties based on their backgrounds. Still, close

cooperation between the people who have part of the experience is indeed essential.

5 DISCUSSION AND CONCLUSION

This study implemented DBL in two kindergarten classes, exploring the challenges and requirements during both design and facilitation of DBL in a Chinese early-childhood education context. Furthermore, information was gathered as a starting point for tool design to support teachers implementing DBL in a more general context. In previous studies, challenges were examined in diverse contexts [36, 40]. Those studies gave ideas about the contextual needs and challenges related to its specific contexts. In general a main reason why teachers perceive more challenges in implementing DBL than traditional kind of teaching methods is because the course is open-ended and transformed from a teacher-centered to student-centered form of teaching. But it is less clear what kind of challenges teachers might face at a more specific and detailed level during implementing DBL. From this study, we find that the challenges participants perceive are highly related to the background knowledge and experience they have. The study has led to a new perspective on analyzing and describing the challenges according to participants' backgrounds based on the teacher's competence model.

We propose a framework, for a teacher's competence model, that is specific to a DBL context, to describe the more detailed challenges for teachers. The framework is an adaptation of the more generic pedagogical content knowledge (PCK) model by Shulman [25]. It includes four blocks: 1) DBL-related knowledge, 2) teaching experience, 3) topic-related knowledge, and 4) material-based knowledge. Those four types of competencies are required during both course design and facilitating phases, and indicate the different competencies needed in implementing DBL compared to in teaching a more general DBL context. The DBL teacher's competence model gives another perspective to think about challenges uncovered in this study. There are two types of participants: 1) The course designers who are experienced with DBL-related knowledge (linked to block 1 in the proposed framework) but lack teaching experience (linked to block 2), and 2) the teachers whose background knowledge is exactly the opposite. The challenges that the participants perceived are mainly related to their lacked experience. The present study that explores the role of the participants' background might seem like an extreme context as problem-based learning approaches are more widely adopted in some regions such as Nordic countries, but teachers in Chinese public schools normally have less experiences with DBL- related approaches because of the differences in educational system.

In the introduction, we presented previous studies in diverse contexts. Using the proposed DBL-specific teachers' competence model, we have a comprehensive perspective to analyze challenges perceived by different teachers and we believe that the DBL teacher's competence model can be used in a variety of contexts. Besides giving more detailed support for reflecting on types of challenges for teachers linked to certain DBL phases, we can also consider how to tailor support for teachers by analyzing teachers' backgrounds based on the competence model. There is a wide variety in the expertise and competences that teachers have. Besides the two 'types' of participants in this study, there are also other variations in what experience teachers have: 1) Having both experience in teaching and DBL, like teachers in a DBL-based school or informal organization, and 2) neither being very experienced in teaching nor in implementing DBL, like some volunteer community teachers, who might know little about DBL, but know about certain topic-based knowledge.

Most materials and tools intending to support teachers implementing DBL provide general support for managing the DBL process, but do not provide tailored support based on the experience of the individual teacher. The DBL teacher competence model can be a valuable model for tool designers to consider how the experience levels for the various elements of the model influence how future tools should provide support. To support teachers, teacher education courses can be personalized and tailored to teachers' competencies. But this does not mean that teachers implementing DBL need to be very experienced in all four aspects in the model, because a DBL course is can also be co-created with people with at least complementary expertise. Thus, cooperation between people with different backgrounds to design the course is essential and discussion about the setup of the course to improve the course is necessary. Because of the diverse backgrounds of different participants, the four types of requirements can be met through cooperation between people who have complementary knowledge or be supported by teacher education courses or supporting tools.

Tables 5 to 8 give a summary of requirements found in this study. When compared to implementing DBL in higher grades, doing DBL in kindergarten has some special aspects to consider: 1) Children have much less focused time, so time management should be considered carefully, and flexible time setting should be taken into consideration, 2) children in kindergarten may have very limited relevant experience. This may differ a lot between children depending on family education. When considering how to make design challenges authentic, it is better to link the challenge to something that is meaningful for them or related to topics already addressed in class.

During the implementation, we also met some challenges regarding course designing and setting: 1) Instead of co-creation between people with different backgrounds, the teacher only gave feedback instead of joining the course design process. From the teacher's perspective, she is not familiar with DBL and in her previous experience, it has worked well that the teaching materials are designed by individual teachers and shared with other teachers to use. 2) Unexpected situations arose during the group ideation phase: the children worked well during individual ideation, but during the group ideation phase they have no experience with this kind of cooperation, so they still draw separately together on the big paper (figure 3). Therefore, it was difficult to continue to the making phase. Two possible solutions can be suggested for this case: 1) Teachers need to give support to each group to ask for their ideas and help them to generate a group design, and 2) taking a more long-term perspective, children can be supported in developing expertise regarding the DBL-related approach or 21st-century skills like cooperation and problem-solving matters. Learning goals regarding 21st century skills to improve children's cooperation skills and creativity need to be seen as long-term goals. Related practices and activities need to be embedded in the daily curriculum to enhance children's ability to design and visualize ideas using different

ways. The cultural context also influences the challenges perceived, such as the high student-to-faculty ratio which means one teacher does not have much time to spend with each student.

As for the stakeholders, the headmasters of kindergarten can play an important role in a formal school to provide support to teachers. Furthermore, potentially, the parents can also play a role to broaden students' capabilities at an early age. Family education can play an important role, especially for young children, and solid connections can be made between the schools and the broader community through parents' involvement [2, 34].

In summary, DBL can play a role in formal kindergarten teaching. However, challenges exist, and tailored support is needed when designing and facilitating DBL. Different kinds of supports can be developed and adjusted to the competencies of the participants using the proposed DBL teacher competence model: 1) Customized teacher education, 2) cooperation between people with complementary competencies, and 3) tools that support certain aspects of competencies. For the kindergartens, which mostly use a more traditional way of teaching like in this case study, more courses and activities that teach 21st century skills are needed.

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